

Technical Details of Red Rover

Head	Details
Operating environment	Lunar Equator and mid-latitudes, sunrise to sunset with night hibernation
Rover mass	80kg
Payload mass	30kg
Average power	120W (400W peak)
Drive speed	10cm/s
Dimensions (L×W×H)	1.4×1.4×1.7m
Features	Night survival, autonomous roving, passive thermal control, shoulder-driven skid steer, passive rocker suspension, composite chassis, 3D-HD cameras, 4x telephoto zoom camera, direct to Earth communication
Services available	Payload delivery, exclusive corporate sponsorship, HD and 3D video, data licences

eyes (cameras) will scan the ground ahead.

Since moon dust is too slippery to permit an accurate reading of distance travelled based on how many times the rover's wheels have turned, the rover's onboard computer will calculate the distance by comparing the changing appearance of surface features as the robot moves. Radiation-hardened components will protect the computers from unfiltered solar and cosmic radiation with which the airless moon is bombarded.

On ground at Pittsburg, Astrobotic team members shall guide Red Rover to interesting features (hopefully, Apollo landing sites) using low-resolution lunar images. The rover will avoid hazards on the moon's surface autonomously. It will beam high-definition videos as blocks of encrypted data, at least one immediately after landing and one later in the mission to meet X Prize requirements. The rover will also send emails, tweets and Facebook posts.

Challenges for Red Rover

A major technical challenge for the team is to ensure that Red Rover survives the extremes of lunar day and night, each of which lasts two Earth weeks. During the two-week lunar night, the temperature on the moon's

surface, where the team plans to land, plummets to -170°C (-274°F). Any component that contains water, such as batteries, would suffer irreparable damage as the water freezes and expands.

The only rover that has survived these extremes of day and night temperatures was Soviet remote-controlled Lunokhods, in 1970s, which relied on radioactive polonium isotope to stay warm. However, Astrobotic and other competing companies do not have access to these materials. To protect Red Rover from the heat of the sun, carbon-filter structures surrounding battery cells conduct heat to the outer surface of the rover. At night, Red Rover will hibernate and will awaken with the sun to fire up non-aqueous lithium-iron-phosphate batteries.

Participating teams

Of the 33 participating teams, Astrobotic has made significant progress, and two purses have already been announced as Astrobotic (the USA) has won US\$ 500,000 in the mobility category and US\$ 250,000 for their imaging sub-system.

In addition to Astrobotic, other teams competing for milestone prizes are Hakuto (Japan), Team Indus (India), Moon Express (the USA) and Part Time Scientists (Germany), as in December 2014.

It is interesting to note that during the same timeframe (by 2016), India and Russia (joint project) and China (which is building its own rover that will use a radio-isotope to stay powered through the lunar night) may land their rovers on the moon's surface.

What can be gained from GLXP

Perhaps the most enduring benefit of GLXP will be to inspire the next gen-

eration of scientists and engineers. Carnegie Mellon's Planetary Robotics Laboratory is entirely dedicated to the moon rover project. They are testing the design for fragmenting metal bolts, an alternative to typical explosive bolts that unhinge the ramps from the spacecraft so that the rover can explore the lunar surface.

Nayaka and a group of undergraduates have prepared a high-speed video camera to record the simulation. After 17.9 seconds of throwing switch, the bolt breaks apart to roll out the rover through the ramp.

For researchers like Douglas Currie, a guaranteed spot for a fixed price on a commercial mission would be a boon. His team wants to place an array of laser ranging reflectometers on the moon to support accurate measurements.

GLXP organisers hope that if they build it, the market will come and, that, developing rovers and getting these on the moon will spur the growth of a new market.

Of course, there has been some skepticism about the competition's viability, and complaints and criticism about the perceived slow pace of progress over the last six years. There is a perceived diminished interest in the moon by NASA, who in 2010 cancelled Constellation Program for a human lunar return and replaced it with plans for human missions to near-Earth asteroids and Mars.

Inspiring, indeed

Even failing to win the prize does not deter some teams, who have visions and business plans that extend beyond the competition.

Richards at SETIcon said, "You know, if another team wins, that is great, too. This is not just about winning the prize. It is about creating a whole new industry. We believe in a long-term vision of opening up moon's resources for the benefit of humanity." ●